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Finite difference-embedded UNet for solving transcranial ultrasound frequency-domain wavefield

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ABSTRACT:

Transcranial ultrasound imaging assumes a growing significance in the detection and monitoring of intracranial lesions and cerebral blood flow. Accurate solution of partial differential equation (PDE) is one of the prerequisites for obtaining transcranial ultrasound wavefields. Grid-based numerical solvers such as finite difference (FD) and finite element methods have limitations including high computational costs and discretization errors. Purely data-driven methods have relatively high demands on training datasets. The fact that physics-informed neural network can only target the same model limits its application. In addition, compared to time-domain approaches, frequency-domain solutions offer advantages of reducing computational complexity and enabling stable and accurate inversions. Therefore, we introduce a framework called FD-embedded UNet (FEUNet) for solving frequency-domain transcranial ultrasound wavefields. The PDE error is calculated using the optimal 9-point FD operator, and it is integrated with the data-driven error to jointly guide the network iterations. We showcase the effectiveness of this approach through experiments involving idealized skull and brain models. FEUNet demonstrates versatility in handling various input scenarios and excels in enhancing prediction accuracy, especially with limited datasets and noisy information. Finally, we provide an overview of the advantages, limitations, and potential avenues for future research in this study. © *2024 Acoustical Society of America*. https://doi.org/10.1121/10.0025391

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I. INTRODUCTION

In contrast to medical imaging modalities like computed tomography (CT) and magnetic resonance imaging (MRI), transcranial ultrasound imaging offers the benefits of being non-invasive, portable, cost-effective, and highly repeatable. It has unique advantages and values in detecting and monitoring intracranial lesions and cerebral blood flow. Due to the fact that the infant skull has not yet fully developed, ultrasound imaging can directly observe the intracranial situation through the fontanelle, making ultrasound imaging a more promising method.

In recent years, there have been some excellent works in the field of solving transcranial wavefields,[1–6] which essentially involve solving partial differential equations (PDEs). Aubry *et al.*[7] proposed a series of numerical benchmarks related to transcranial ultrasound simulation, introduced 11 modeling tools, and systematically compared the numerical results obtained by different tools. The commonly used grid-based numerical techniques, such as finite difference (FD) and finite element methods, have been effectively utilized in the domain of medical ultrasound imaging.[8–11] However, due to the grid-based concept, the computational complexity of these methods increases exponentially when dealing with high-frequency or high-dimensional problems, and even suffers from the curse of dimensionality.[12] Another important issue to consider is the error introduced by discretization. Grid-based methods solve discretized PDEs through time stepping. When the grid size is insufficient to meet the desired modeling resolution, discretization errors may lead to inaccurate results.[13] In addition, the performance of FD method is not satisfactory when facing with irregular boundary problems. Moreover, the use of FD method to solve anisotropic acoustic wave equations often leads to shear wave artifacts linked to the Laplacian operator, particularly when sources are situated in the anisotropic region.[14] Another aspect that cannot be ignored is that when the physical parameters of the medium change slightly, these numerical methods need to be iteratively restarted, which brings some inconvenience to practical problem solving. Therefore, exploring an alternative approach to wavefield solving is of considerable importance.

Neural networks (NNs) and their variants, such as the convolutional neural network (CNN) and recurrent neural network, have always been regarded as universal tools for solving various types of problems. Given a sufficient amount of training data and a NN architecture with enough hidden layers and neurons, they can effectively grasp both linear and non-linear connections between the input and

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output. Many studies have demonstrated that NNs can be used as an alternative tool to obtain solutions to PDEs.[15–17] Despite achieving great success, there are still some limitations with NNs. First, NNs perform poorly on small sample datasets. NNs require a large amount of training data to achieve excellent models in a vast parameter space. If the training dataset is too small, it usually leads to overfitting or underfitting. However, obtaining a large amount of data is not always easy for practical problems. Second, there are issues with handling imbalanced datasets. When the sample data of one class is much greater than that of other classes, NNs may tend to predict this class, leading to poor prediction results for other classes. Third, there is a lower requirement for model interpretability. NNs are often viewed as "black boxes" and their internal decision-making process is difficult to explain. Due to the lack of constraints in the learning process from physical knowledge such as the conservation of energy and momentum, NNs can easily produce solutions that deviate significantly from the truth. In response to the above situation, physics-informed neural network (PINN) and its derivatives, as new paradigms for solving PDEs, have attracted people's attention and been widely studied.[18–20] The core idea of PINN is to leverage the automatic differentiation technique to encode the governing equations and boundary conditions, along with other prior knowledge, within the error function of a NN. By enforcing the physical laws as constraints, the network iterations are guided and constrained according to the underlying physics. The effectiveness of PINN in modeling wave equations has been demonstrated in both time and frequency-domains.[4,21–24] However, PINN can only perform iterative calculations on the same model each time. When we modify any physical parameter in the model, PINN needs to be retrained. PINN has higher requirements for the initial and boundary conditions. To take advantage of it, more accurate initial and boundary conditions need to be provided, otherwise unreliable computational results will be obtained. Furthermore, before applying PINN, it is necessary to reformulate the PDE into a suitable form. Sometimes, the application of PINN may be limited when dealing with complex PDEs. Therefore, developing modeling techniques that are easy to implement, rely less on datasets, and can be applied to different models could potentially bring significant benefits in the field of transcranial ultrasound.

In addition to the above discussion, there is another issue that cannot be ignored. It is computationally costly to obtain the wavefield solution in time domain, especially for multiple sources.[25] Additionally, there is a significant need for memory to store the wavefield information. Compared with time-domain problems, solving the frequency-domain wavefield can markedly reduce computational costs for an equivalent number of sources. Moreover, when using frequency-domain data for inversion, it is only necessary to choose a few key frequency points for fitting in order to obtain inversion results that are close to the actual model.[26] This can greatly reduce the amount of data and computational effort required. Furthermore, frequency-domain inversion is generally more stable and accurate compared to time-domain inversion, due to the weaker correlation between different frequency components and the better resistance to disturbance factors such as noise.[27,28] Through the application of methods like full waveform inversion, the solution of the extensive impedance matrix derived from the wave equation (Helmholtz equation) yields Green's functions that can be utilized for model inversion.[14]

In this paper, we introduce a framework called FD-embedded UNet (FEUNet) to solve the frequency-domain transcranial ultrasound wavefield corresponding to the Helmholtz equation. The proposed approach chooses UNet, a widely used fully CNN consisting of an encoder and a decoder, as the data-driven framework. The input of the network is a two-dimensional (2D) brain velocity model, containing clots of varying sizes and positions. The output is the frequency-domain wavefield corresponding to the model, encompassing both the real and imaginary parts. FEUNet is inspired by the idea of PINN and adds physical constraints, the PDE error calculated from the governing equation, to the data-driven error function. Both components jointly constrain the network iterations, which is different from purely data-driven methods. We acquire numerical wavefield solutions by employing an optimal 9-point FD wave equation operator,[29] and use these results as the reference solutions to evaluate the performance of both the pure data-driven method (UNet) and the proposed method (FEUNet). We compared the performance of UNet and FEUNet using the idealized skull model and 2D brain model, respectively, and also discussed the impact of different numbers of samples and signal-to-noise ratio (SNR) conditions on the approaches. The results demonstrate that the proposed method can accelerate the iteration process of the network and improve the prediction performance, especially for scenarios with limited samples and noisy data.

II. METHODOLOGY

A. Wave equation in frequency domain

Due to the intricate structure of the skull and its intricate interaction with ultrasound waves, accurately modeling the transmission of transcranial ultrasound presents a formidable challenge. The coupling of solid and liquid components, the presence of multiple layers, and the porous nature of the skull all exert a significant influence on the behavior of ultrasound waves as they traverse it. Furthermore, the skull's high velocity relative to the internal tissues adds an additional layer of complexity to the modeling process. However, when we reasonably neglect the effects of shear wave, nonlinearity, and acoustic absorption,[1,30–33] process in frequency domain can be represented as

$$\nabla^2 p(\boldsymbol{x},\omega) + \omega^2 m p(\boldsymbol{x},\omega) = s(\boldsymbol{x}_s,\omega), \tag{1}$$

where $p(\boldsymbol{x},\omega)$ is the complex pressure wavefield, $s(\boldsymbol{x}_s,\omega)$ is the source function, ω stands for the angular frequency, $\boldsymbol{x} = \{x, y\}$ represents the spatial coordinate in the Cartesian

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coordinates, and $\boldsymbol{x}_s$ represents the spatial location of the acoustic source. We define $m = 1/v^2$ as the squared slowness with v denoting the velocity and $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2$ as the Laplacian operator. Equation (1) is alternatively labeled as the Helmholtz equation. In numerical simulations, we often use absorbing boundary conditions[34] or perfectly matched layer (PML) techniques[35] to simulate infinite space without introducing interference from reflected waves, ensuring the accuracy and stability of the calculations. Once PML is incorporated into a 2D bounded rectangular area, the Helmholtz equation can be represented by[25,36]

$$\frac{\partial}{\partial x}\left(M\frac{\partial p}{\partial x}\right) + \frac{\partial}{\partial y}\left(N\frac{\partial p}{\partial y}\right) + K\omega^2 mp(\boldsymbol{x},\omega) = s(\boldsymbol{x}_s,\omega), \tag{2}$$

where the coefficients M, N and K are respectively equal to P_y/P_x, P_x/P_y, and P_xP_y. The values of P_x and P_y can be determined by

$$P_{x,y} = 1 - \frac{i\zeta_{x,y}}{\omega}. \tag{3}$$

The purpose of introducing $\zeta_{x,y}$ is to attenuate and absorb wave energy within the boundary of PML. Beyond the boundaries of PML, both ζ_x and ζ_y have a value of zero. Inside the PML, ζ_x and ζ_y are expressed as

$$\zeta_{x,y} = 2\pi a_0 f_0 \left(\frac{d_{x,y}}{D_{PML}}\right)^2, \tag{4}$$

where f_0 represents the dominant frequency of the source function and a_0 is a constant coefficient equal to 1.79.[36] D_{PML} stands for the thickness of PML layers; d_x and d_y represent the horizontal and vertical distances from points (x, y) within the PML to the boundaries of the PML region, respectively. By combining Eqs. (2)–(4), we can implement the PML technique for wavefield solution on a 2D brain model.

B. The architecture of UNet

We chose UNet as the main framework for the NN to solve the transcranial ultrasound frequency-domain wavefield. Its structure is shown in Fig. 1. UNet was originally proposed by Ronneberger *et al.* in 2015 for medical image segmentation[37] and has since been applied to wavefield simulation problems. In contrast to the original structure, we changed the corresponding parameters such as the depth of the network and the value of padding to make it more suitable for the specific problem addressed in this paper. The input of the network is the 2D brain velocity model, including blood clots of different sizes, positions, and velocities (following Gaussian distribution). The output of the network consists of two channels, representing the real and imaginary components of the frequency-domain wavefield corresponding to the input model. The network architecture consists of an encoder (referred to as the downsampling process, shown in the upper part of Fig. 1) and a decoder (referred to as the upsampling process, shown in the lower part of Fig. 1), connected through multiple skip connections. In the encoder, each downsampling step includes two 3×3 convolutional layers, followed by a Leaky ReLU activation function and a 2×2 maxpooling layer. Within the decoder, every upsampling stage consists of a 2×2 transposed convolutional layer, followed by a concatenation operation and two consecutive 3×3 convolutional layers.

In NNs, the output q^i of the ith layer is calculated from the output of the previous one,

$$q^i = f^i\left(W^i q^{i-1} + b^i\right), \quad i = 1, 2, \ldots, L, \tag{5}$$

where W^i is the weight matrix of the ith layer, b^i is the bias vector, q^{i-1} represents the output of the $(i-1)$th layer,

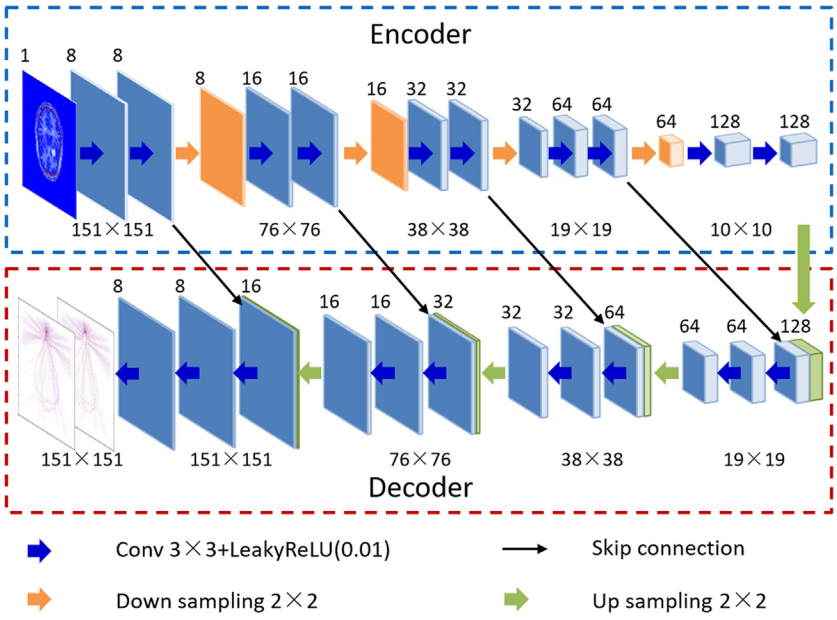


FIG. 1. (Color online) The network structure of UNet.

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f^i is the activation function. The training process of the NN is an iterative process, with each iteration updating and optimizing the weight matrix W and bias matrix b in the network, so that the output of the network gradually approaches the labeled value. The error function, in this work, is the mean squared error (MSE), commonly used to measure the error between the model's output and the actual label, with the optimization process striving to minimize it:

$$MSE_{Data} = \frac{1}{N_D}\sum_{i=1}^{N_D}(p - p^*)^2, \tag{6}$$

where p and p^* represent the labeled solutions and the predicted values of the NN. MSE_{Data} here refers to the data-driven MSE in UNet. N_D is the number of collocation points used to calculate MSE_{Data}. In all the works, this error term is calculated using all grid points. The Adam optimizer is used to minimize the error function.[38]

C. The architecture of FD-embedded UNet (FEUNet)

Incorporating domain knowledge as constraints during the model optimization process and implementing these constraints by modifying the framework of the NNs is a feasible way to improve network performance.[39] Based on this idea, algorithms such as PINN, TgNN (theory-guided neural network)[40] have been proposed and successfully applied in various fields. These algorithms share almost identical computation principles, incorporating governing equations as regularization terms into the total error function of NNs, ensuring the predicted results are close to physical mechanisms. These algorithms can only target the same input each time, that is, each iteration optimizes the wavefield of the same model. In this paper, we propose the FD-embedded UNet (FEUNet) framework, and the algorithm framework is shown in Fig. 2. The biggest difference between the proposed algorithm and PINN is that it uses the idea of adding governing equation constraints, but the input of the network is different models. It combines the advantages of data-driven and physics-driven modeling approaches. FEUNet can make predictions on various different inputs and use physical constraints to enhance the robustness and interpretability of the prediction results. This may be very useful for many practical problems. Compared to the input of PINN (coordinates x, y, and time t for time-domain problems, coordinates x, y, and angular frequency ω for frequency-domain problems), the input of FEUNet is a 2D brain model containing blood clots with different sizes, positions, and velocities. Therefore, automatic differentiation cannot be used to obtain partial derivatives and calculate the PDE error. In the numerical simulations of this paper, the training set is generated by an optimal 9-point FD wave equation operator. Therefore, we use the concept of it to calculate the partial derivatives here.

Assume that an arbitrary 2D bounded domain is divided into a uniform grid. Let $p_{x,y}$ represents the pressure wavefield at coordinate (x, y) and Δd stands for the node spacing on the grid. According to the 9-point FD method, the Laplacian operator can be expressed as

$$\nabla^2 p_{x,y} = a\nabla^2_{(0)}p_{x,y} + (1-a)\nabla_{(45)}{}^2 p_{x,y}, \tag{7}$$

where

$$\nabla^2_{(0)}p_{x,y} = \frac{p_{x+1,y} + p_{x-1,y} - 4p_{x,y} + p_{x,y+1} + p_{x,y-1}}{\Delta d^2}, \tag{8}$$

$$\nabla^2_{(45)}p_{x,y} = \frac{p_{x+1,y+1} + p_{x-1,y+1} - 4p_{x,y} + p_{x+1,y-1} + p_{x-1,y-1}}{2\Delta d^2}. \tag{9}$$

The value of constant a is 0.5. Meanwhile, the FD approximation of the collocation point p of the mass acceleration term $\omega^2 mp$ in Eq. (1) can be represented as the following linear combination:

$$\begin{aligned} p_{x,y} = {} & bp_{x,y} + c(p_{x+1,y} + p_{x-1,y} + p_{x,y+1} + p_{x,y-1}) \\ & + d(p_{x+1,y+1} + p_{x-1,y+1} + p_{x+1,y-1} + p_{x-1,y-1}), \end{aligned} \tag{10}$$

where $b + 4c + 4d = 1$. The optimized coefficients given by Jo *et al.*[29] are $b = 0.6248$, $c = 0.09381$, and $d = -0.00004$. Substitution of Eqs. (7)–(10) into Eq. (1) gives the following equation:

$$\begin{aligned} & a\frac{p_{x+1,y} + p_{x-1,y} - 4p_{x,y} + p_{x,y+1} + p_{x,y-1}}{\Delta d^2} + (1-a) \\ & \times \frac{p_{x+1,y+1} + p_{x-1,y+1} - 4p_{x,y} + p_{x+1,y-1} + p_{x-1,y-1}}{2\Delta d^2} \\ & + \omega^2 m\Big[bp_{x,y} + c(p_{x+1,y} + p_{x-1,y} + p_{x,y+1} + p_{x,y-1}) \\ & + \frac{1-b-4c}{4}(p_{x+1,y+1} + p_{x-1,y+1} + p_{x+1,y-1} \\ & + p_{x-1,y-1})\Big] = s. \end{aligned} \tag{11}$$

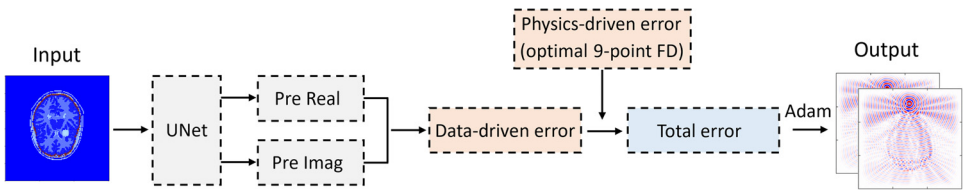


FIG. 2. (Color online) The workflow diagram of FEUNet.

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Equation (11) is the FD formulation of Eq. (1). By substituting the predicted frequency-domain wavefield obtained from the network into Eq. (11) and selecting N_P collocation points for MSE calculation, we can obtain the physics-driven error term MSE_{Phy}. The total error of the network is obtained by weighted summation of data-driven error and physics-driven error,

$$MSE_{total} = \lambda_{Data} MSE_{Data} + \lambda_{Phy} MSE_{Phy}. \tag{12}$$

The selection of weights for λ_{Data} and λ_{Phy} is contingent on the particular circumstances. One general solution is to make each error term on the same order of magnitude. This weight configuration scheme can make the network easier to converge. Yet in fact, there is no fixed answer to the selection of the weights. Sometimes increasing or decreasing the initial weights by a factor of ten may yield improved iterative results. When the sample of the dataset decreases, it is advisable to increase the weight parameters of physical constraint terms appropriately. We obtain the most appropriate answer through trials and errors.

III. NUMERICAL SIMULATIONS

In this section, we will share the results of implementing the proposed method on the idealized skulls and the 2D brain model, respectively. Both the training and testing sets were obtained from the optimal 9-point FD wave equation operator. For all cases, the source function was implemented using a Ricker wavelet. To partially alleviate the singularity inherent in point sources, isotropic Gaussian sources with a variance of 3 were employed as a surrogate.[25] During frequency-domain wavefield modeling, the Ricker wavelet was converted from the time domain into a set of complex values. The Gaussian source was multiplied by the appropriate Fourier representation of the Ricker wavelet complex value for a specific frequency. The batch size for all examples is set to 16.

Before conducting numerical simulations, it is necessary to test the validity of the physical constraints by calculating the PDE errors before and after network training. Here, we take the 2D brain model as an example (a detailed introduction about this model will be provided later). As shown in Fig. 12 in the Appendix, the PDE errors for 500 random points after network training are much smaller than before. This indicates that the direction of the physical constraint is consistent with that of the network iteration, proving the effectiveness of the physical constraint. After the completion of network training, we validate the prediction performance of the network on the testing set. We use the relative 2-norm error ℓ_2 to characterize the magnitude of prediction errors. The relative 2-norm error ℓ_2 is calculated as follows:

$$\ell_2 = \frac{\|p_{pre} - p_{ref}\|_2}{\|p_{ref}\|_2}. \tag{13}$$

A. Performance comparison on idealized skull models

These examples are trained using a dataset of sound speed distributions obtained from idealized models of cranial structures. The training dataset was modified from the original dataset mentioned in Ref. 1. Figure 3 displays several velocity distribution examples derived from idealized skull models. As the excitation source, we selected the Ricker wavelet with a central frequency of 500 kHz, positioned at coordinates (131.25 mm, 248.75 mm). The idealized skull model is randomly generated and has a hollow convex structure resembling an ellipse. Each example has a constant thickness, but the thickness varies between different examples, ranging from 5.5 mm to 27 mm. All the examples have the same velocity of 2820 m/s, with a fixed background velocity of 1500 m/s. This configuration ensures that the sound speed contrast between the background and the idealized skull aligns with the speed contrast between soft tissue and the human skull. Each example has a size of 1000×1000 grid points, with a normalized grid spacing of 0.25 mm. The thickness of each boundary in PML is 25 grids. The training set consists of 512 different idealized skull models, while the testing set contains 100 distributions. This example is to demonstrate the performance of the proposed method in

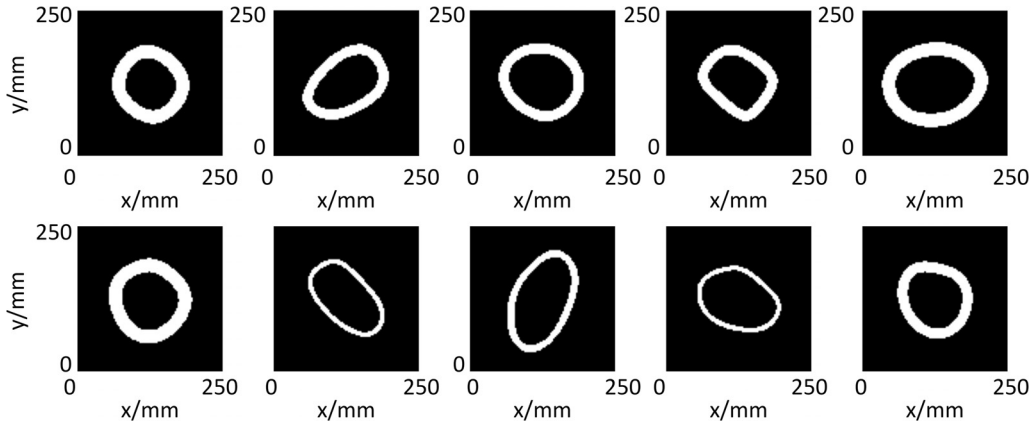


FIG. 3. Examples of the velocity distributions based on idealized skulls. The size of each case is 1000×1000, with a spacing of 0.25 mm.

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processing skull models.[1,39] The number of sampling points for calculating the PDE error is 2000, accounting for 0.2% of the total number of the allocation points.

Training was conducted using both UNet and FEUNet models, and the error curves are depicted in Fig. 4. From the error curve, it can be observed that FEUNet achieves lower training error (0.01822) compared to UNet (0.05260). In addition, due to the incorporation of physical constraints, FEUNet exhibits faster iteration speed, as evidenced by requiring fewer iterations to reach the same error value. The relative 2-norm errors ℓ_2 for the 100 examples in the testing set are shown in Fig. 5. Overall, the errors for FEUNet model are consistently smaller than those for UNet.

Figure 6 presents four examples of wavefield computations. For simplicity, we only compared the real part of the wavefield. From the predicted wavefield, we can observe that UNet is capable of predicting most of the wavefield information. We can also infer the approximate shape of the input from the wavefield. However, there is a significant error of wavefield information after the ultrasound passes through the idealized skull model. In contrast, with the incorporation of physical constraints, FEUNet is better able to predict the subtle wavefield information, especially in the regions near the skull where the wavefield becomes more complex due to sudden changes in velocity. The differences between the references and the predictions provide a more intuitive reflection of this result, where it is evident that the FEUNet exhibits significantly lower errors compared to UNet. These results demonstrate that the physical constraints improve the predictive capability of the network, making it easier for the network to capture the detailed information of the wavefield.

B. The impact of different numbers of observations on predictions

These examples are trained based on the 2D brain model. The 2D brain model is derived from the multimodal imaging-based detailed anatomical (MIDA) model. This model presents a highly detailed segmented cranial structure

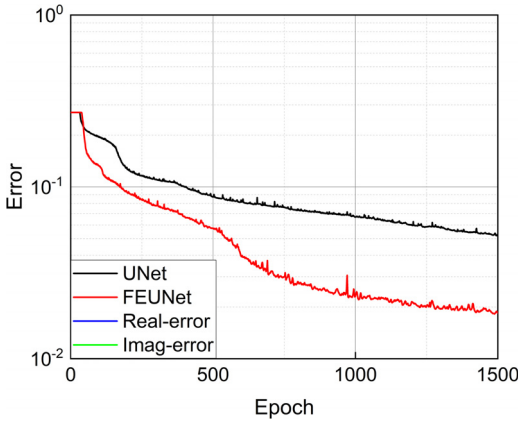


FIG. 4. (Color online) The network error of UNet and FEUNet for idealized skull model.

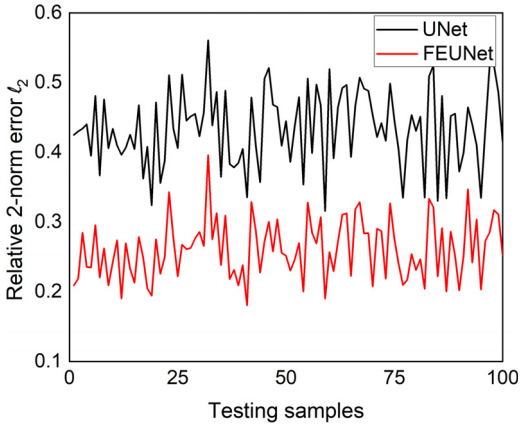


FIG. 5. (Color online) The relative 2-norm error ℓ_2 of UNet and FEUNet for idealized skull models in testing set.

through in-depth processing and analysis of high-quality MRI, magnetic resonance angiography, and diffusion tensor imaging scan data.[41] The MIDA model provides various tissues and corresponding acoustic parameters such as velocity and density for a 3D brain model. These parameters may not be complete and missing values need to be retrieved from literature. Cutting the MIDA model with a 0.5 mm interval yields 465 slices, from which we selected one slice [as shown in Fig. 7(a)] to build the 2D brain model [as shown in Fig. 7(c)]. The prevailing consensus posits that the outer structure of the brain model is immersed in a medium of water, exhibiting a velocity of 1500 m/s. We can also obtain the brain model parameters from CT or MRI images using the relationship between Hounsfield units (HU) and acoustic parameters. This theory is widely recognized, and detailed calculation formulas can be found in Refs. 29 and 30. Angla *et al.*[42] provided a comprehensive summary of the calculation of brain acoustic parameters. We added blood clots with different locations, sizes, and velocity distributions to the original 2D brain model, and used them as the velocity models in both the training and testing sets. These blood clots are all circular in shape, with radii ranging from 8 to 20 mm. The centers of the clots are located within a circle with a radius of 30 mm centered at the midpoint. The velocity distribution of blood clots adheres to a Gaussian distribution. We randomly generated a maximum value not exceeding 1700 m/s and a minimum value not less than 1600 m/s. Subsequently, the velocity distribution between the maximum and minimum values was generated following a Gaussian distribution. Figure 7(d) shows one example of the 2D brain model with a clot. The velocity profile at the black line is shown in Fig. 7(b). We do not have a fixed number of samples in the training set, as we will discuss the influence of the number of samples on the proposed algorithm. In the testing sets, we regenerated blood clots with different positions, sizes, and velocity distributions. Unlike the training sets, the velocity distribution of clots in the testing set includes fixed values of 1600 and 1700 m/s, as well as velocities following a Gaussian distribution (as described above). The size of each input in the examples is 151×151,

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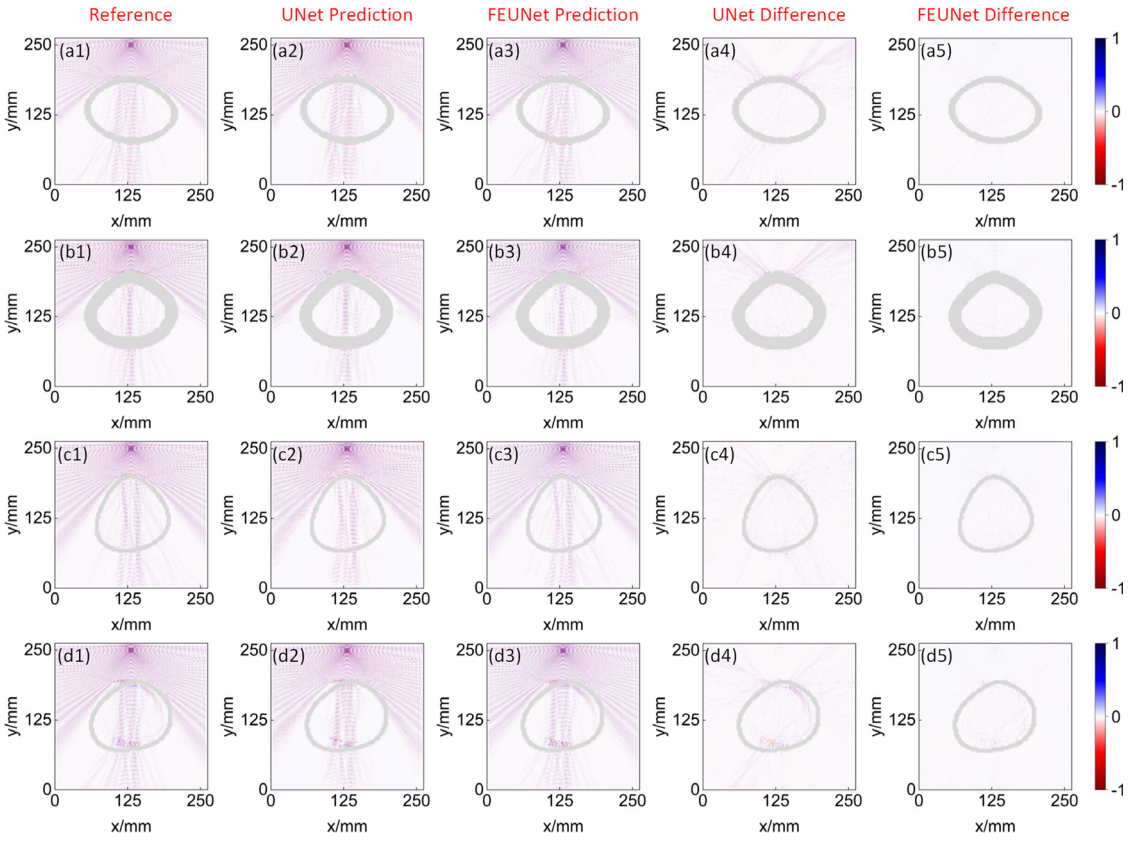


FIG. 6. (Color online) Four examples of simulations using idealized skull distributions randomly selected from the testing set. The idealized skull models are covered on the wavefields and the wavefields have been normalized. In each case, the reference solution is obtained from the optimal 9-point FD wave equation operator.

and the output is $151 \times 151 \times 2$, where "2" represents the real and imaginary parts. The grid spacing is 2 mm.

Figure 7 depicts the 2D brain model (with and without a blood clot) and their corresponding transcranial ultrasound frequency-domain wavefields. FEUNet has the ability to convert governing equations into constraints and effectively leverage domain knowledge to optimize network iterations. In theory, it possesses greater learning capability and requires less data compared to purely data-driven models (UNet). To evaluate the effectiveness of the proposed method, we compared the performance of FEUNet and UNet on different numbers of training sets, ranging from 64 to 512. The excitation source with a main frequency of 300 kHz is located at (150 mm, 256 mm). The thickness of each boundary in PML is 20 grids. For all the cases, a random selection of 500 points (2% of the total allocation points) are employed to calculate the PDE error. Independent replication experiments were conducted to avoid the influence of randomness on model evaluation. After training, we obtained the predictions of the two methods on the same testing set and compared the results with a reference solution obtained by the optimal 9-point FD calculations. Figure 8 shows the error curves and errors of FEUNet and UNet under different numbers of training sets, as well as their errors of the real part of frequency-domain wavefield compared to the reference solution. We can see from the error curves that for different numbers of training samples, the training errors of FEUNet are always smaller than UNet. Additionally, FEUNet requires far fewer iterations to reach the same error value as UNet, which reflects FEUNet's faster iteration speed compared to UNet. The error curves for the real and imaginary parts in the PDE error also demonstrate the iteration of the network. Meanwhile, the errors of the wavefields shows that FEUNet has smaller errors than UNet, particularly in cases where the training dataset is less extensive. This is quite meaningful in practical problems, as a large amount of training data is often difficult to obtain in practical situations. The use of physical constraints can reduce the network's dependence on data and achieve better network prediction results in the case of a small amount of data. It should be noted that we adopt different weight configuration schemes for different amounts of training data. The initial solution involves establishing a ratio between the two weights, ensuring that the data-driven error and physics-driven error have values within the same order of magnitude. When the dataset is small, it is advisable to increase the weight of the physical constraints (e.g., by a factor of 10) appropriately. The specific weight configuration scheme was obtained through trials and errors. Furthermore, we calculated the relative 2-norm errors ℓ_2 for both methods on 100 testing sets and drew a box plot to compare the distribution of errors, as shown in Fig. 9. It is very clear that for all cases, the errors of FEUNet are

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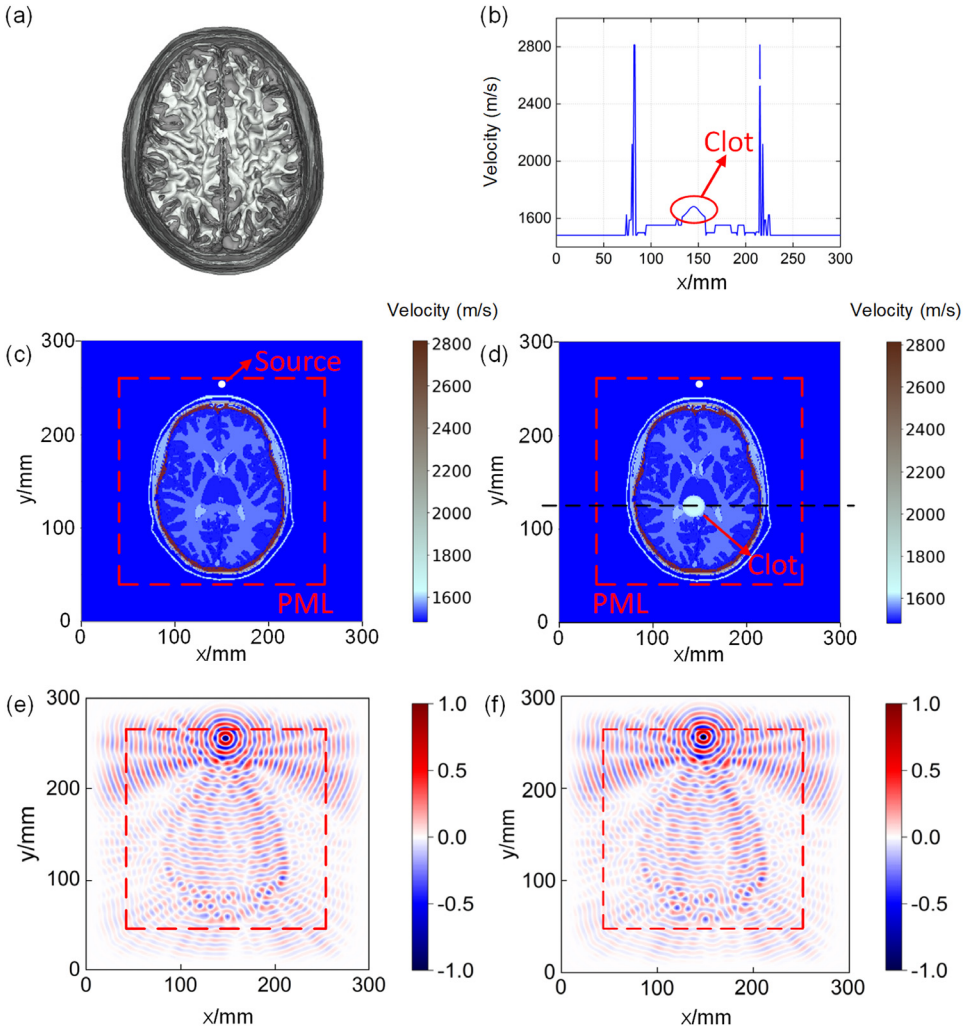


FIG. 7. (Color online) The 2D brain model (with and without a blood clot) and their corresponding transcranial ultrasound frequency-domain wavefields. (a) A slice from MIDA; (c) 2D brain model transformed from (a); (d) 2D brain model with a clot; (b) the velocity profile at the blood clot in (d); (e) the frequency-domain wavefield of (c); and (f) the frequency-domain wavefield of (d). The red dashed lines represent the PML. The wavefields have been normalized.

lower than that of UNet. When the number of training samples is 64, the error of UNet sharply increases to a large value, indicating that the prediction of the wavefield by UNet has almost completely failed. The increase in FEUNet error is relatively small, almost equal to the error of UNet in the case of 128 samples, which proves that physical constraints reduce the network's dependence on the training data. In addition, it can be seen from the concentration of errors that FEUNet has a smaller variance value and better applicability to different inputs.

C. The impact of different SNR conditions on predictions

Within this section, we explore the influence of noise on the effectiveness of the proposed algorithm. Random noise, following a normal distribution with a mean of zero, was introduced to the complex wavefields in the training set of the 2D brain model. The training set consists of 256 samples, while the testing set contains 100 samples. The signal-to-noise ratio (SNR) ranges from 10 to −10 dB. For each SNR condition, both algorithms were trained multiple times and the best results on the testing set were selected for comparison. The comparison of relative 2-norm error ℓ_2 of UNet and FEUNet under different SNR conditions is shown in Fig. 10. As the SNR decreases, the errors of both methods gradually increase. However, the error values of FEUNet are always lower than those of UNet, and the upward trend is also smaller. It is more evident that the performance of the network is improved with the addition of physical constraints under low SNR conditions. In the real world, observations inevitably contain both noise and genuine values. Purely data-driven models struggle to differentiate between authentic data and noise, making them susceptible to substantial interference. However, because true values themselves conform to physical mechanisms, the incorporation of domain-specific knowledge empowers FEUNet to effectively discriminate data noise, thereby enhancing the model's resilience to noise interference to a certain extent.

IV. DISCUSSION

A. Advantages of the proposed method

FEUNet is a NN-based forward encoder-decoder model. It draws inspiration from PINN and incorporates governing equations into the error function to constrain network iterations and enhance network representation capabilities. This improves the network's expressive power and generalization ability, allowing it to predict subtle acoustic information,

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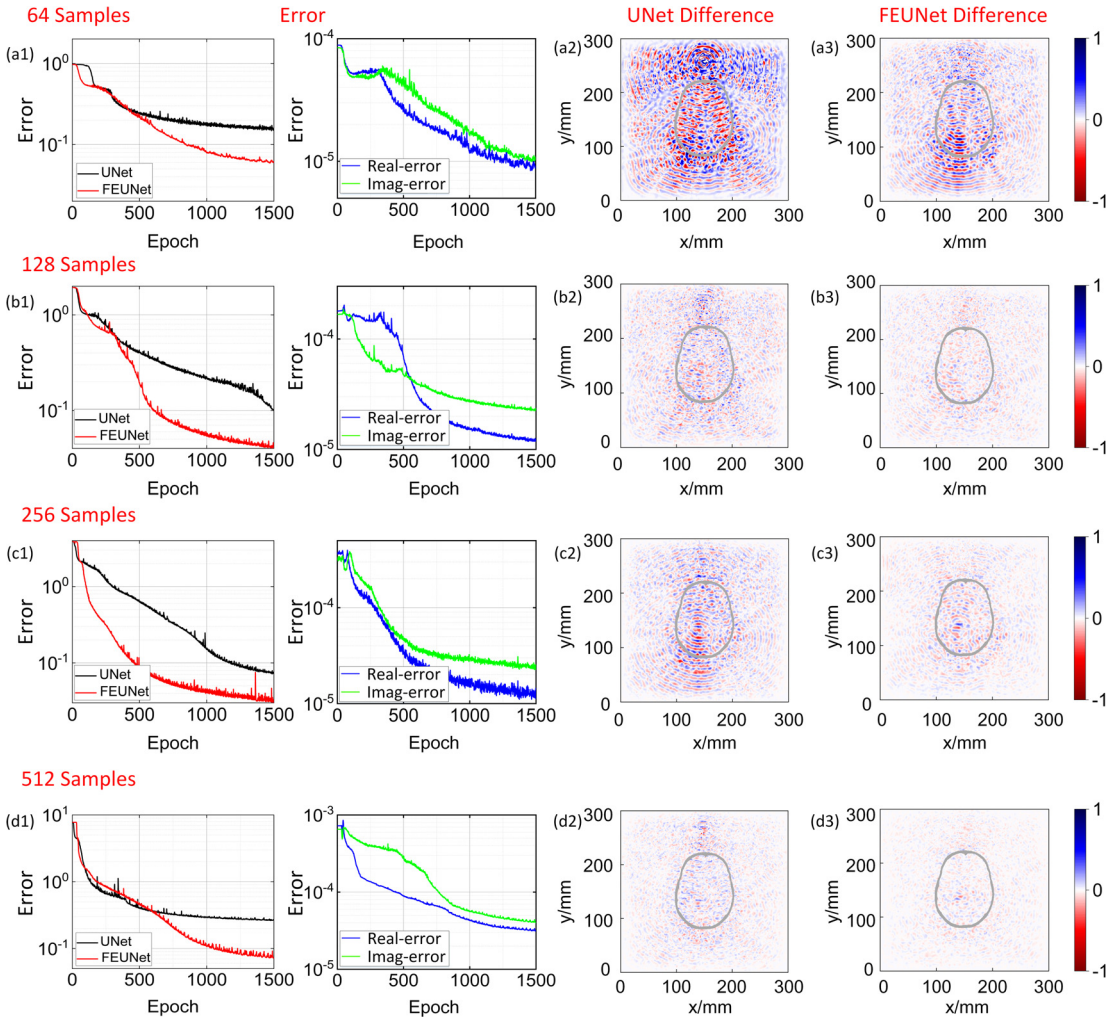


FIG. 8. (Color online) The network error of UNet and FEUNet under different numbers of training sets, as well as the differences between their respective predicted solutions and the reference solutions. The reference solution is obtained from the optimal 9-point FD wave equation operator. The skull models are covered on the wavefields and the wavefields have been normalized.

reducing the network's dependence on the dataset and improving its noise resistance. It also retains the framework of traditional NNs, allowing it to be applicable to different input models, which can be more challenging for PINN. Therefore, it combines the advantages of multi-model training and improving network performance. It is important to note that when using the PINN framework for frequency-domain modeling of the 2D brain model in this paper, it is necessary to preprocess the velocity model by applying smoothing techniques. This is because PINN may encounter difficulties when dealing with boundaries with abrupt velocity changes. Here, we use the method proposed in Ref. 25 (PINN for solving frequency-domain wavefields) for comparison and explanation. The input for PINN is the 2D brain model with a size of 151×151 and a grid spacing of 2 mm. The output is the corresponding frequency-domain wavefield, which consists of both the real and imaginary parts. As mentioned previously, the optimal 9-point FD wave equation operator is employed to generate the reference solution. We select the Ricker wavelet with a frequency of 150 kHz as the excitation source, located at (150 mm, 256 mm). Figure 11 displays the prediction results of PINN. For simplicity, only the real part of the wavefield is presented here. Figures 11(b) and 11(c) show the real part of the wavefield predicted by PINN for the original model, as well as the difference compared to the reference solution. It is evident that PINN has failed to predict the corresponding wavefield, particularly the information surrounding the brain,

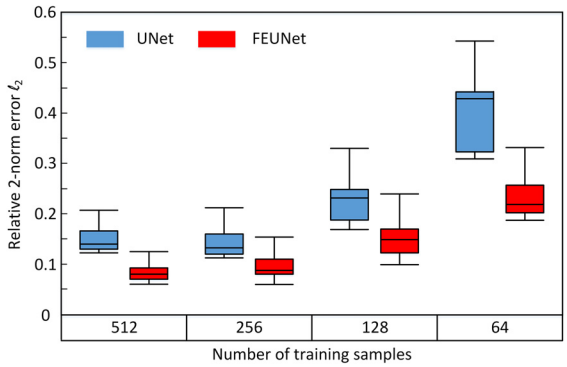


FIG. 9. (Color online) The relative 2-norm error ℓ_2 of UNet and FEUNet under different numbers of training sets.

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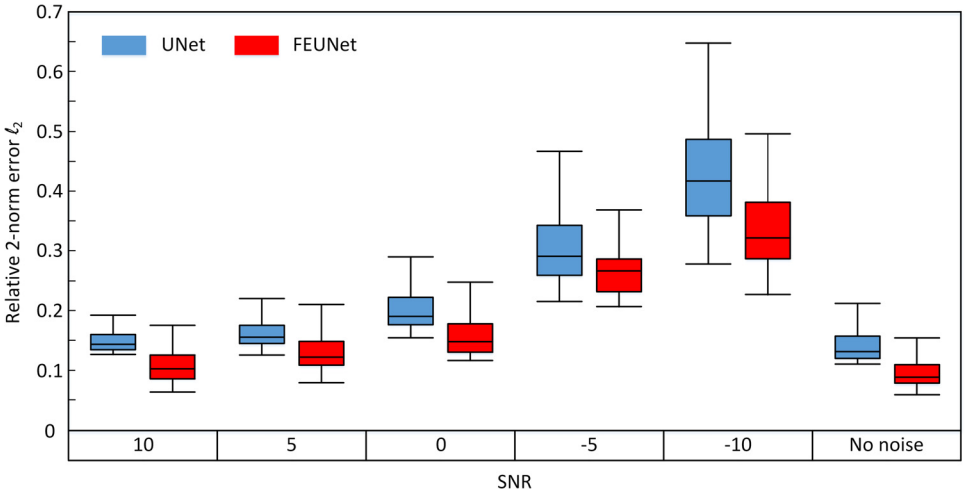


FIG. 10. (Color online) The relative 2-norm error ℓ_2 of UNet and FEUNet under different SNR conditions.

which is almost negligible. After the original model is smoothed, as shown in Fig.11(d), PINN is capable of accurately obtaining the solution. This is also supported by the difference shown in Fig.11(f). By comparing the two sets of results, we can infer that PINN requires prior smoothing preprocessing of the velocity model when predicting the frequency-domain wavefield. In this example, we have chosen a lower frequency of 150 kHz. The reason for selecting a lower frequency is that PINN struggles to handle high-frequency problems effectively. Increasing the frequency requires a deeper network with a larger number of neurons. Despite the low frequency, this example is sufficient to illustrate the limitations of PINN in predicting frequency-domain wavefields and its inability to handle models with significant velocity fluctuations. In contrast, FEUNet is not constrained by the velocity model. In addition, we compared the time required for different methods (FEUNet, UNet, and FD) to obtain wavefields for the same testing set (sample size of 100), as shown in Table I in the Appendix. The reason for the difference in the runtime of the two models is the disparity in their sizes. The data size of the idealized skull model is 210×210, while the 2D brain model is sized at 150×150. This also indicates, from a side perspective, that the increase in model size has a significant impact on the runtime of the FD method.

B. Limitations, solutions, and future directions

FEUNet achieves the proximity of prediction results to the underlying physical laws by incorporating the governing equations into the network's error function as regularization terms. This approach serves as a soft constraint, assessing the weighted average consistency between the model predictions and the physical laws. Although the regularization terms constrain the network's iterations to some extent and ensure that the prediction results comply with the physical laws on average, they cannot guarantee strict adherence to the physical laws at every single point (which is a limitation of the penalty method). In practice, it is possible to encounter notably unrealistic physical predictions that significantly deviate from the governing equations. In contrast to soft constraints, hard constraints refer to projecting the predictions of the NN onto a hyperplane constructed by governing equations, ensuring that the results on each constraint patch strictly adhere to the physical constraints.[39,43] As a result, hard constraints are more efficacious than soft constraints and can further enhance network performance. In theory, by applying hard constraints, the model can achieve higher prediction accuracy and stronger noise resistance while requiring much less data. However, the implementation of hard constraints is always more challenging than soft constraints. Currently, the research on incorporating hard constraints into deep learning is still at its early stages. Employing the implementation of hard constraints to improve the performance of FEUNet would be a promising avenue for the future work.

In this study, we utilize a 300 kHz central frequency signal to stimulate the wavefield of the 2D brain model. While this frequency is relatively low in comparison, opting for a higher frequency such as 500 kHz, the FD method would require the grid to be divided into multiple times in order to obtain a dispersion-free wavefield. However, in this case, the number of points contained in the frequency-domain wavefield increases exponentially. Training a network on this dataset would require a larger number of parameters, which would be very time-consuming and may even exceed the computational capabilities available at present. In fact, FEUNet can be relatively easily extended to handle problems involving other frequencies and models. When dealing with different problems, we need to modify the expression of the corresponding physical constraints using the concept of FD method (as the governing equation may be different), as well as the weight parameters.

Due to the embedding of physical constraints, additional PDE error calculation is performed during network training, which unavoidably increases the computational cost. However, only 2.2% of the collocation points are added for the calculation of PDE errors. More importantly, as compensation, FEUNet offers higher prediction accuracy compared to UNet. It is capable of capturing more detailed wavefield information, reducing reliance on the dataset, and enhancing its ability to resist noise interference. In addition, from the error curve we can observe that FEUNet has a faster iteration speed compared

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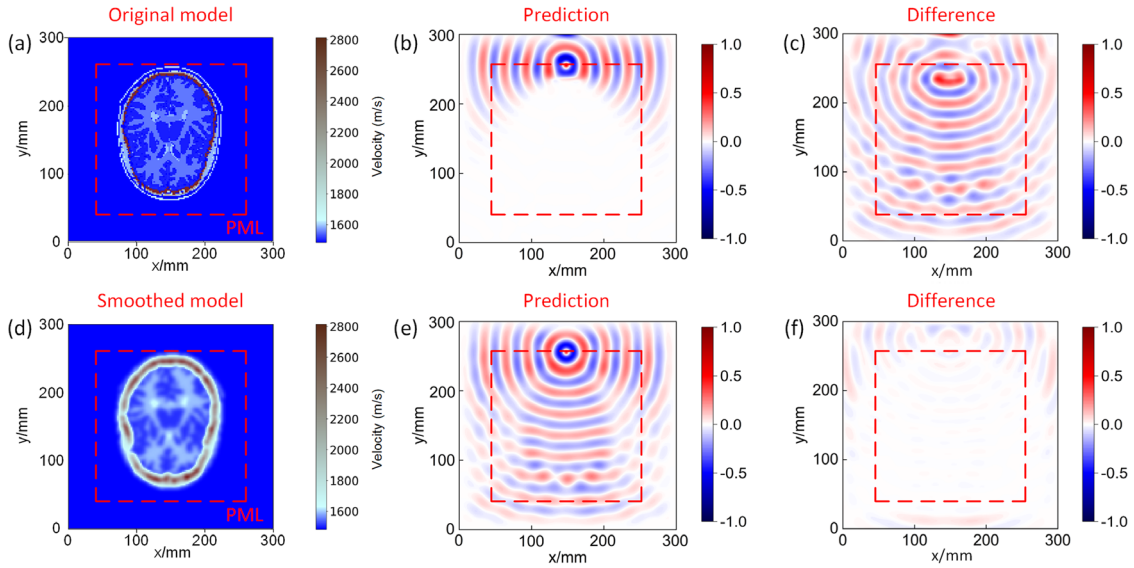


FIG. 11. (Color online) The prediction results of PINN before and after smoothing. The red dashed lines represent the PML. The wavefields have been normalized.

to UNet. It requires fewer iteration steps to reach the same error value. Compared to PINN, FEUNet exhibits better generalization, as it can be applied to different models once the network is well trained, without limitations on the velocity model. For complex physical problems, the modeling process often necessitates considering multiple constraint conditions to approximate the behavior of real systems. In such situations, the design of the error function and the adjustment of its weights become crucial to ensure the accurate fitting of data and compliance with physical constraints. However, in the face of these challenges, the network architecture we proposed shares the same training mechanism, which allows us to avoid cumbersome computations of large matrix inversions. This method provides an efficient and feasible solution for solving complex physical problems. By judiciously selecting constraint terms and error function weights, we can achieve precise control over the system's behavior, leading to more reliable and accurate simulation results. It is important to emphasize that this method is not limited to specific fields of physics. Its generality and scalability pave the way for in-depth exploration in physical modeling and offer new possibilities for interdisciplinary studies, including fields like computer vision, natural language processing, and biomedical research. Drawing inspiration from this network architecture's principles, we can tackle complex real-world problems in these domains. Similarly, Chatillon *et al.*[44] suggested an alternative approach by substituting the computationally expensive direct solver with a surrogate model constructed from a database of simulation results. This method facilitates rapid and precise simulation of the 3D pressure field and thermal damages caused by HIFU in near real-time. In addition, Granados *et al.*[45] proposed a physics-based, explainable, fast, and reliable meta-model aiming at providing a controllable data generation tool. Our future work will consider learning and utilizing this model to obtain more accurate solutions for multiple physical fields. Overall, this approach presents a promising direction for both physical modeling and interdisciplinary research. Its potential applications hold value in pushing the boundaries of scientific knowledge and technological advancements.

Finally, this approach can be extended to 3D and more complex wavefields to obtain more accurate transcranial ultrasound wavefields. It requires inputting 3D labeled data and incorporating constraints from the 3D wave equation when applying this approach. For more complex scenarios, such as considering solid-liquid coupling and the porous nature that closely resembles the actual conditions of the brain, as well as accounting for attenuation and absorption effects during wave propagation, we would need more accurate governing equations and physical constraint error terms. Additionally, applying appropriate boundary conditions and initial conditions would also enhance the accuracy of the network in predicting the wavefield. It is also a valuable recommendation to achieve adaptive configuration of hyperparameters under soft constraint conditions. The combination of the data-driven framework and the incorporation of physical constraints makes it easier for this method to learn complex wavefield information. The implementation of this approach in transcranial imaging applications suggests its potential applicability to other complex physical systems as well.

V. CONCLUSION

In this paper, we propose an FD-embedded UNet (FEUNet) framework to solve transcranial ultrasound frequency-domain wavefields. This method is based on a data-driven framework while incorporating the concept of PINNs, which embeds the governing equations into the error function of the NN. Unlike PINNs, which take coordinates as inputs to the network, this method takes different velocity models as inputs and produces corresponding frequency-domain wavefields as outputs. It possesses the advantages of both data-driven and physical constraint approaches. It can be applied to

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different models while ensuring that the learned mapping relationship incorporates the underlying physical laws. The performance of this method has been validated using the idealized skull models and 2D brain model with blood clots. The results demonstrate that it can achieve higher prediction accuracy and capture finer details of the wavefields compared to UNet. The embedding of physical constraints reduces the network's reliance on data and enhances its robustness against noise. Future works include applying this approach to 3D brain imaging, incorporating additional physical constraints such as boundary conditions and initial conditions to improve network performance, adopting self-adjusting penalty weights, applying the concept of hard constraints and considering complex viscoelastic anisotropic media or fluid-saturated porous media. Before applying it to imaging, further works are needed to extend the method to more general model configurations, such as scenarios with multiple excitation sources.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY

The data that support the findings of this study are openly available in JASACode_FEUNet at https://figshare.com/s/78ddfcc45fbbef8ee5dc.

APPENDIX

The PDE error before and after training for 500 random points is shown in Fig. 12.

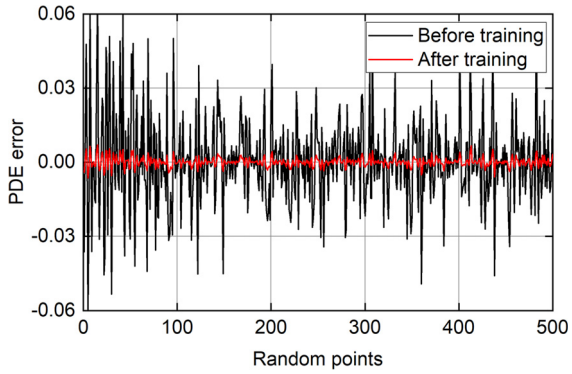


FIG. 12. (Color online) The PDE error before and after training for 500 random points.

TABLE I. Comparison of time required for different methods (unit: seconds).

	FEUNet	UNet	FD
Idealized skull model	17.8807	18.2621	74.2894
2D brain model	11.0852	11.1698	38.9433

The comparison of the time required for obtain wavefields for the same testing set (sample size of 100) using different methods is shown in Table I.

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